GSE 518
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Problem Set 3

You must turn in problems: 3, 5 and 6. However, you are strongly encouraged to work on all the problems before the midterm, since the material they cover will be part of the midterm. Please let me know if you have any questions.

1. Let X be a continuous random variable with probability density function (pdf) $f_X(x)$. Is it true that $f_X(x)$ can only take values between 0 and 1? If yes, show it. If not, provide an example (*Hint:* You may want to look at Figures 2.3.1 or 2.3.2 from CB).

2. (a) Problem 1.52 in CB.
   (b) Prove that $f(x) = 2x/(k(k+1))$ for $x = 1, 2, \ldots, k$ and zero otherwise is a probability mass function. (*Hint:* $\sum_{x=1}^{k} x$ is a well known arithmetic series. You may want to find out what that sum equals to.)
   (c) Problem 1.47 part (c).

3. Let X be a random variable with probability density function

$$f(x) = \begin{cases} 0.2 & \text{if} \quad -1 \leq x \leq 0 \\ 0.2 + cx & \text{if} \quad 0 < x \leq 1 \end{cases}$$

   (a) Find the value of c that makes $f(x)$ a pdf.
   (b) Find the cdf $F_X(x)$.
   (c) Find the probability: $\Pr(0 \leq X \leq 0.5)$.

4. Let X be a continuous random variable with pdf given by

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2},$$

   where $-\infty < x < \infty$, $-\infty < \mu < \infty$ and $\sigma > 0$. In this case X is said to follow a normal or Gaussian distribution, usually written as $N(\mu, \sigma^2)$. Let $Y = aX + b$, where a and b are constants, and $a \neq 0$. Use the pdf of X to derive the pdf of Y. What is the distribution of Y? Let $Z = (X - \mu)/\sigma$, what is the distribution of Z?

5. Problem 2.3 in CB.

6. Problem 2.1 in CB parts (b) and (c).

2.5 Exercises

2.1 In each of the following find the pdf of Y. Show that the pdf integrates to 1.

~~(a) $Y = X^3$ and $f_X(x) = 42x^5(1-x)$, $0 < x < 1$~~

(b) $Y = 4X + 3$ and $f_X(x) = 7e^{-7x}$, $0 < x < \infty$

(c) $Y = X^2$ and $f_X(x) = 30x^2(1-x)^2$, $0 < x < 1$

(See Example A.0.2 in Appendix A.)

2.2 In each of the following find the pdf of Y.

(a) $Y = X^2$ and $f_X(x) = 1$, $0 < x < 1$

(b) $Y = -\log X$ and X has pdf

$$f_X(x) = \frac{(n+m+1)!}{n!\,m!}x^n(1-x)^m, \quad 0 < x < 1, \quad m, n \text{ positive integers}$$

(c) $Y = e^X$ and X has pdf

$$f_X(x) = \frac{1}{\sigma^2}xe^{-(x/\sigma)^2/2}, \quad 0 < x < \infty, \quad \sigma^2 \text{ a positive constant}$$

2.3 Suppose X has the geometric pmf $f_X(x) = \frac{1}{3}\left(\frac{2}{3}\right)^x$, $x = 0, 1, 2, \ldots$. Determine the probability distribution of $Y = X/(X+1)$. Note that here both X and Y are discrete random variables. To specify the probability distribution of Y, specify its pmf.

2.4 Let λ be a fixed positive constant, and define the function $f(x)$ by $f(x) = \frac{1}{2}\lambda e^{-\lambda x}$ if $x \geq 0$ and $f(x) = \frac{1}{2}\lambda e^{\lambda x}$ if $x < 0$.

(a) Verify that $f(x)$ is a pdf.

(b) If X is a random variable with pdf given by $f(x)$, find $P(X < t)$ for all t. Evaluate all integrals.

(c) Find $P(|X| < t)$ for all t. Evaluate all integrals.

2.5 Use Theorem 2.1.8 to find the pdf of Y in Example 2.1.2. Show that the same answer is obtained by differentiating the cdf given in (2.1.6).

2.6 In each of the following find the pdf of Y and show that the pdf integrates to 1.

(a) $f_X(x) = \frac{1}{2}e^{-|x|}$, $-\infty < x < \infty$; $Y = |X|^3$

(b) $f_X(x) = \frac{3}{8}(x+1)^2$, $-1 < x < 1$; $Y = 1 - X^2$

(c) $f_X(x) = \frac{3}{8}(x+1)^2, -1 < x < 1$; $Y = 1 - X^2$ if $X \leq 0$ and $Y = 1 - X$ if $X > 0$

2.7 Let X have pdf $f_X(x) = \frac{2}{9}(x+1)$, $-1 \leq x \leq 2$.

(a) Find the pdf of $Y = X^2$. Note that Theorem 2.1.8 is not directly applicable in this problem.

(b) Show that Theorem 2.1.8 remains valid if the sets $A_0, A_1, \ldots, A_k$ *contain* $\mathcal{X}$, and apply the extension to solve part (a) using $A_0 = \emptyset$, $A_1 = (-1, 1)$, and $A_2 = (1, 2)$.

2.8 In each of the following show that the given function is a cdf and find $F_X^{-1}(y)$.

(a) $F_X(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 - e^{-x} & \text{if } x \geq 0 \end{cases}$

(b) $F_X(x$

(c) $F_X(x$

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